La Scuola d'Italia / Dr. Huson / IB Math 11
19 October 2026

Nickname:

2 Homework: Slope, Lines and Systems of Equations

1. The points $A(-2, 1)$ and $B(4, 4)$ lie on a straight line.
   (a) Find the gradient (slope) of the line AB.
   (b) Find the equation of the line AB in the form $y = mx + c$.
   (c) Find the x-intercept and the y-intercept of the line AB.
   (d) The point $C(10, k)$ also lies on the line AB. Find the value of k.

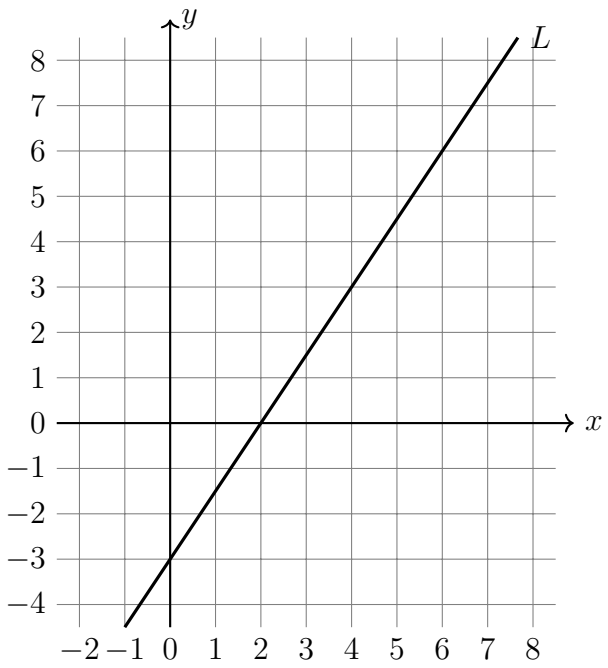
2. The line L_1 has equation $2x + 3y = 12$.
   (a) Find the x-intercept and the y-intercept of L_1.
   (b) Find the gradient of L_1.
   (c) The line L_2 is perpendicular to L_1 and passes through the point $(4, 1)$. Find the equation of L_2.
   (d) The line L_3 is parallel to L_1 and has y-intercept -2. Write down the equation of L_3.

3. Solve each system of equations. Show your working.

(a) $3x + 2y = 16$
$x - y = 2$

(b) $2x + 3y = 7$
$5x - 2y = -11$

4. The line L is shown on the grid.



(a) Write down the equation of L in the form $y = mx + c$.

(b) The line M has equation $x+y = 7$. Draw the line M on the grid.

(c) Mark the point where L and M intersect on the grid. Write down its coordinates as an ordered pair.

(d) Use algebra to solve the system formed by the equations of L and M.